



Smart Telescope Sort

User Manual · Version 1.1

macOS · Free and open source

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Organize TIFF, FITS and JPG captures from several smart-telescope brands into one year / object library. The app does not connect to the telescope: you copy sessions onto your Mac first, choose that Captures folder, and the app recognizes the folder layout for you.



Smart Telescope Sort

Multi-brand Captures → Targets

HOW IT WORKS

- Copy the telescope's dated session folders into Captures. The layout is detected; Targets (year) folders are skipped.
- Pick a year, a month and the file types to move: TIFF, JPG/JPEG, FITS/FIT or All.
- Review file plan lists every file with its object, date and target folder.
- Optional Zip or Tarball backup. You name it, and a progress window shows it running.
- Sort moves the ticked file types into Targets (year)(object). A newer copy replaces an older one.
- Files already in Targets are marked Duplicate. You're asked Yes/No before they go to the Trash.
- Emptied capture folders are removed. A folder still holding unticked images asks Yes/No first.

Open user manual (PDF)

Source code on GitHub

Astronomy Observation Planner

Mac App Store

Smart Telescope Planner

iPhone & iPad App Store

Nothing moves until you press Sort eligible files.

CAPTURE LIBRARY

Ready to review

Dated session folders - detected

SOURCE CAPTURES

/Users/Shared/Astronomy/Captures

ORIGINAL TARGETS

/Users/Shared/Astronomy/Targets

PROCESSING TARGETS

/Users/Shared/Astronomy/Vespera Processing

BACKUP STORAGE

/Users/Shared/Astronomy/Sort Backup

DESTINATION

Choose a year and month

Objects decode into Targets (year)(DSO) for the detected capture layout.

Year All years Month All mon... Refresh preview

FILES TO MOVE

☒ TIFF ☒ JPG / JPEG ☒ FITS / FIT ☐ All

BACKUP

Tarball (.tar.gz) → /Users/Shared/Astronomy/Sort Backup

5 capture rows found

Destination shows where TIFF/FITS files will move

Capture folder	Year	Month	DSO object	Img folders	Files	Status	Formats	Destination
2026-09-10_22-15-03_observation_M31	2026	09	M31	1	3	3 new	FITS, TIFF	Targets 2026/M31
2026-09-12_01-40-22_observation_NG...	2026	09	NGC7023	2	3	3 new	FITS, TIFF	Targets 2026/NGC7023
2026-09-14_07-20-55_plan_Demo_Night	2026	09	M27	1	1	1 new	FITS	Targets 2026/M27
2026-09-14_07-20-55_plan_Demo_Night	2026	09	NGC7000	2	2	2 new	TIFF	Targets 2026/NGC7000
2026-09-18_23-05-11_observation_M31	2026	09	M31	1	2	2 new	TIFF	Targets 2026/M31

5 folders ready to inspect 11 image files to sort

Replace only when newer. Existing target files are kept unless the source is newer.

11 files eligible: 11 new moves, 0 newer replacements.

Review file plan Sort eligible files

Preview ready for Dated session folders (TIFF, FITS / FIT) — no files have been moved.

The main window: the folder layout is detected, and every capture folder is listed with its object, file count and destination before anything moves.

1. Assumptions (read this first)

- **You already moved files off the scope** (USB, FTP, or Wi-Fi). The app never talks to Vaonis, Seestar, DWARF, or Origin hardware.
- **One brand per Captures folder.** The layout is detected from the folder you choose. Do not mix Seestar albums and Vaonis dated sessions in the same Captures folder.
- **You choose the file types:** TIFF, JPG / JPEG, FITS / FIT, or All. TIFF and FITS are ticked by default; JPG is usually for viewing and sharing.
- **Apple Photos is not a source.** Export to a Captures folder if images only live in Photos.
- **Every brand uses the same library layout:** Targets {year}/{object}/ inside your Original Targets folder.
- **Year** comes from dated folder names when present; otherwise from the newest image file's modification date.
- **Object name** is decoded from observation / album / object folder names.
- **Replace only when newer.** If Targets already has the same file name, the newer copy wins. Identical or older copies are marked Duplicate and you are asked about them.
- **Targets {year}** folders are never scanned as sources and are never removed.

1.1 What the app does not do

- It does not stack, stretch, or post-process images.
- It does not support Unistellar in this version.
- It does not invent missing object names — if a folder has no usable name, the object label may be the parent folder name.

2. Your four folders

The first time the app opens it asks where three library folders live. Tick **Save this as default** to keep the choice; leave it unticked to use it for this session only. Each folder has a **Change...** button in the main window.

- **Source Captures** — where you copied the telescope's sessions. Choose it with the ... button. The folder layout is detected each time the preview refreshes.
- **Original Targets** — holds your Targets {year} folders of sorted originals. If it is not set, Targets folders stay inside Source Captures.
- **Processing Targets** — where you process objects, one folder per object (for example Vespera Processing). The file plan shows a green check when an object already has a processing folder.
- **Backup Storage** — where backups are written before a sort.

```
{Original Targets}/Targets {year}/{object}/  
example: .../Targets/Targets 2026/M31/  
example: .../Targets/Targets 2026/M42/
```

If the year folder does not exist yet, the app offers to create Targets {year} first. No files move when it is created.

3. Detected folder layouts

Telescope names identify folder layouts only. Smart Telescope Sort is an independent app and is not affiliated with, endorsed by, or created by those manufacturers. The detected layout is shown under the title, for example *Dated session folders · detected*.

Layout	Telescopes	What the folders look like
Dated session folders	Vespera, Stellina	2026-09-10_22-15-03_observation_M31
Object album folders	S30, S30 Pro, S50	M31/, NGC253/ (or a USB dump of albums)
Session folders	DWARF 3, II, mini	20260907_DWARF3_session/M33/
Object-and-date folders	Origin Mark II	M31_2026-09-05/

4. How each telescope gets files onto the Mac

Offload is done outside this app. Below: how vendors typically expose files and what the app expects after you copy them into Captures.

4.1 Vespera / Stellina

- **Off the scope:** FTP the **User/** dated folders. Newer models also support USB-C transfer. Multi-night: stop each night so the project saves, then copy each night's dated folder.
- **In Captures:** keep dated names unchanged, e.g. 2026-09-21_11-43-55_observation_M31 or ..._plan_My_plan with nested 01-observation-... and 01-images-....

```
User/      (on telescope)
  2026-09-10_22-15-03_observation_M31/
    01-images-initial/
      img-0001.tiff
      img-0002.fits
→ copy whole folder into Captures → Sort → Targets 2026/M31/
```

4.2 S30 / S30 Pro / S50

- **Off the scope:** USB cable (drive often shows **MyWorks** / object folders), Wi-Fi file share, or app export of FIT.
- **In Captures:** object albums with .fit / .fits inside (e.g. M31/, or a USB dump containing several albums).

```
MyWorks/
  M31/
    ...fit
  NGC253/
    ...fits
→ copy albums (or the dump) into Captures → Sort → Targets {year}/M31/
```

4.3 DWARF 3 / II / mini

- **Off the scope:** USB mass storage (appears as a disk) and/or FTP (often ftp://192.168.88.1 on the telescope's Wi-Fi).
- **In Captures:** keep each session folder intact. The object may be a nested folder (e.g. .../M33/) under a dated session.

```
Astronomy/<session>/
  M33/
```

```
stacked.fits
stacked.tiff
sub_0001.fits
```

→ copy session into Captures → Sort → Targets {year}/M33/

4.4 Origin Mark II

- **Off the scope:** enable **Save Raw Images**, then copy via USB stick (FAT32/exFAT) from the app File Manager, or FTP to a computer.
- **In Captures:** folders named object + date, e.g. M31_2026-09-05/ with .fits. A parent USB dump with several object + date folders also works.

```
M31_2026-09-05/
  raw_0001.fits
  raw_0002.fits
```

→ copy into Captures → Sort → Targets 2026/M31/

5. Step by step

- Copy sessions off the telescope into a Captures folder.
- Open **Smart Telescope Sort** and answer the folder questions (first launch only).
- Set **Source Captures** (...) to that folder. The layout is detected.
- Pick a **Year** and **Month** (or All), and tick the **Files to Move**: TIFF, JPG / JPEG, FITS / FIT, or All.
- Choose a **Backup**: Off, Zip (.zip), or Tarball (.tar.gz).
- Press **Review file plan** to inspect every file, its target folder and status.
- Press **Sort eligible files**. Name the backup, watch it run, then confirm **Sort now**.
- Answer the Yes / No questions about duplicates and finished folders.
- Browse Targets {year}/{object} for stacking and archive.

File plan

Dated session folders

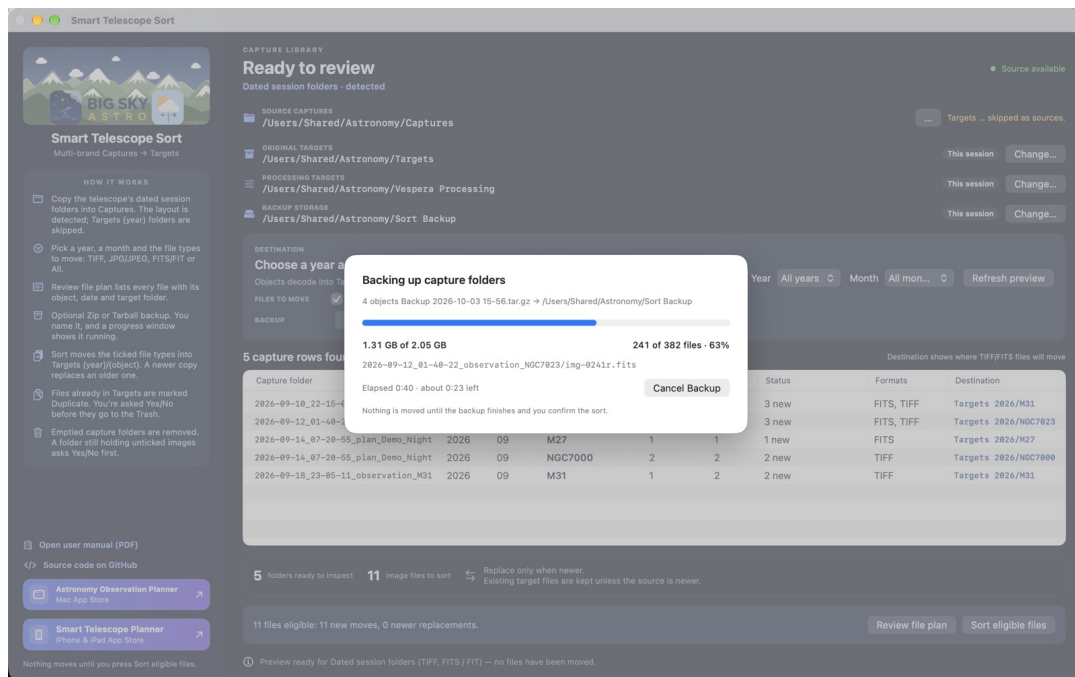
4 capture folder(s) - 11 files: 11 new · 0 newer · 0 duplicate. Nothing has been moved yet.

DSO / name	Image file	Date	Target folder	Processing	Status
2026-09-10_22-15-03_observation_M31 3 new					
M31	img-0001.tiff	2026-09-21 17:08	Targets 2026/M31	✓ M31	New
M31	img-0002.tiff	2026-09-21 17:08	Targets 2026/M31	✓ M31	New
M31	img-0003.fits	2026-09-21 17:08	Targets 2026/M31	✓ M31	New
2026-09-12_01-40-22_observation_NGC7023 3 new					
NGC7023	img-0001.tiff	2026-09-21 17:08	Targets 2026/NGC7023	—	New
NGC7023	img-0008.tiff	2026-09-21 17:08	Targets 2026/NGC7023	—	New
NGC7023	img-0009.fits	2026-09-21 17:08	Targets 2026/NGC7023	—	New
2026-09-14_07-20-55_plan_Demo_Night 3 new					
M27	img-0001.fits	2026-09-21 17:08	Targets 2026/M27	✓ M27	New
NGC7000	img-0001.tiff	2026-09-21 17:08	Targets 2026/NGC7000	✓ NGC7000	New
NGC7000	img-0010.tiff	2026-09-21 17:08	Targets 2026/NGC7000	✓ NGC7000	New

Review file plan lists every file grouped by capture folder, with its target folder, processing folder and status. Nothing has moved yet.

6. Backups

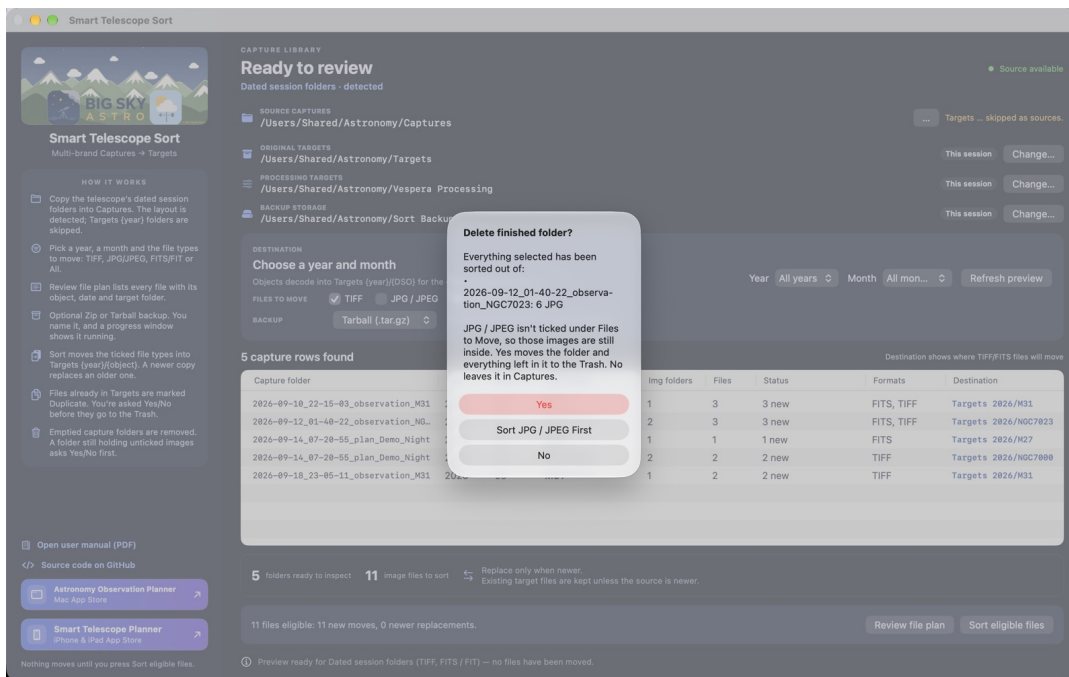
- **Zip or Tarball** — chosen in the **Backup** menu. Before it starts you are asked to name it; the suggestion is the objects plus the date and time, e.g. M31 M27 Backup 2026-10-03 15-27.tar.gz. A number is added if the name is already used.
- A progress window shows the size and file count done, time elapsed and time left. **Cancel Backup** stops it and removes the partial archive; nothing is moved.
- When the backup finishes you are asked to confirm the sort. If it fails, nothing is moved.
- **Backup Off** — the app still offers **Back up first...**, which copies the capture folders into a named folder in Backup Storage, or **Sort without backup**.
- Archives are written fast (light compression) using the archiver built into macOS. If it is missing or a backup fails, **Help** → **Free Zip & Tarball Apps** lists free alternatives.



A named Tarball backup running, with Cancel Backup.

7. Duplicates and finished folders

- **Duplicates** — files already in Targets (identical or older) are marked *Duplicate*. After the sort you are asked **Delete duplicates?** Yes moves them to the Trash; the Targets copies are not touched. No leaves them in Captures.
- **Emptied capture folders** are removed after a successful sort.
- **Finished folders** — when a capture folder is sorted but still holds images of a type you did not tick (for example JPG), you are asked **Delete finished folder?** **Yes** moves the folder and everything left in it to the Trash, **Sort JPG / JPEG First** ticks that type and sorts it, and **No** leaves the folder in Captures.
- Removed folders go to the Trash where the drive supports it.



Delete finished folder? — Yes, sort the remaining type first, or No.

8. Safety rules

- Nothing moves until you confirm the sort.
- A destination file is replaced only when the source is newer.
- Targets {year} is never scanned as a capture source.
- The Captures folder and Targets year folders are never removed during clean-up.
- Backups copy whole capture folders; they never modify Targets.
- Every deletion asks Yes / No first.

9. Free and open source

Smart Telescope Sort is free, and its source code is on GitHub: github.com/rderry/Smart-Telescope-Sort. Open it from **Source code on GitHub** in the sidebar or **Help** → **Source Code on GitHub**. The app first asks for credit, then opens the repository when you click **Continue**.

If you change it and give it away, please give credit to BigSkyAstro: include the BigSkyAstro logo and a link to bigskyastro.com. Credit line: “Based on Smart Telescope Sort by BigSkyAstro — <https://bigskyastro.com>”.

10. Help and support

- Privacy policy: <https://bigskyastro.com/privacy>
- App page: <https://bigskyastro.com/macOS/smart-telescope-sort>
- Support: <https://bigskyastro.com/feedback/smart-telescope-sort> · support@bigskyastro.com

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